1. Problem Statement:

A company wants to manage employee salaries with different policies. Create classes

Employee, Designer, and Tester.

• Each subclass should calculate salary differently:

o Designer: salary = base + 10% of base

o Tester: salary = base + 5% of base + bonus

• Implement constructor chaining and method overriding

```
// Base class
class Employee {
   int id;
   String name;
   double baseSalary;

   // Constructor
   Employee(int id, String name, double baseSalary) {
      this.id = id;
      this.name = name;
      this.baseSalary = baseSalary;
   }

   // Method to calculate salary (will be overridden)
   double calculateSalary() {
      return baseSalary;
   }

   void display() {
      System.out.println("ID: " + id + ", Name: " + name + ", Salary: " +
calculateSalary());
```

```
    }
}

// Subclass Designer
class Designer extends Employee {

    Designer(int id, String name, double baseSalary) {
        super(id, name, baseSalary); // Constructor chaining
    }

    // Method Overriding
    double calculateSalary() {
        return baseSalary + (0.10 * baseSalary);
    }
}

// Subclass Tester
class Tester extends Employee {
    double bonus;

    Tester(int id, String name, double baseSalary, double bonus) {
        super(id, name, baseSalary); // Constructor chaining
        this.bonus = bonus;
    }

    // Method Overriding
    double calculateSalary() {
        return baseSalary + (0.05 * baseSalary) + bonus;
    }
```

```
}

// Main class
public class Company {
    public static void main(String[] args) {
        Designer d = new Designer(101, "Sejal", 40000);
        Tester t = new Tester(102, "Khaleesi", 35000, 5000);

        System.out.println("Employee Salary Details:");
        d.display();
        t.display();
    }
}
```

2. Problem Statement:

Create a Swing GUI applicaϴ on that accepts a string. Provide buΣ ons to:

1. Count consonants
2. Display string in lowercase
3. Show string length

Input Format:

- Input string via JTextField

Output Format:

- Display result in JLabe

```
import javax.swing.*;
import java.awt.event.*;
```

```
public class SimpleStringApp extends JFrame implements ActionListener {
    JTextField tf;
    JLabel lbl;
    JButton b1, b2, b3;

    SimpleStringApp() {
        tf = new JTextField(15);
        lbl = new JLabel("Result here");
        b1 = new JButton("Consonants");
        b2 = new JButton("Lowercase");
        b3 = new JButton("Length");

        add(new JLabel("Enter String:"));
        add(tf);
        add(b1);
        add(b2);
        add(b3);
        add(lbl);

        setLayout(new java.awt.FlowLayout());
        setSize(300, 200);
        setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);

        b1.addActionListener(this);
        b2.addActionListener(this);
        b3.addActionListener(this);
    }
```

```
    public void actionPerformed(ActionEvent e) {
        String s = tf.getText();
        if (e.getSource() == b1) {
            int count = 0;
            for (char c : s.toLowerCase().toCharArray()) {
                if ("bcdfghjklmnpqrstvwxyz".indexOf(c) != -1)
                    count++;
            }
            lbl.setText("Consonants: " + count);
        } else if (e.getSource() == b2) {
            lbl.setText("Lowercase: " + s.toLowerCase());
        } else {
            lbl.setText("Length: " + s.length());
        }
    }

    public static void main(String[] args) {
        new SimpleStringApp();
    }
}
```

3. Problem Statement:

Input an array of integers. Display:

- Count of odd numbers
- Sum of first and last element
- Numbers divisible by 3

```
import java.util.Scanner;
```

```
public class ArrayPractice {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size: ");
        int n = sc.nextInt();
        int[] arr = new int[n];

        System.out.println("Enter elements:");
        for (int i = 0; i < n; i++)
            arr[i] = sc.nextInt();

        int oddCount = 0, divisibleBy3 = 0;

        for (int num : arr) {
            if (num % 2 != 0)
                oddCount++;
            if (num % 3 == 0)
                divisibleBy3++;
        }

        int sumFirstLast = arr[0] + arr[n - 1];

        System.out.println("Odd numbers count: " + oddCount);
        System.out.println("Sum of first and last: " + sumFirstLast);
        System.out.println("Numbers divisible by 3: " + divisibleBy3);
    }
}
```

// Q1: Accept and Display Student Details

```
import java.util.Scanner;

public class StudentDetails {     public
static void main(String[] args) {

        // Create Scanner object for console input
        Scanner sc = new Scanner(System.in);

        // Input student details
        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Roll Number: ");
int rollNo = sc.nextInt();

        System.out.print("Enter Marks: ");
double marks = sc.nextDouble();

        // Display formatted output
        System.out.println("\n--- Student Details ---");
        System.out.printf("Name       : %s\n", name);
        System.out.printf("Roll Number: %d\n", rollNo);
        System.out.printf("Marks      : %.2f\n", marks);

        // Close the scanner
        sc.close();
```

```
    }
}
```

Q Area Calculator (Using Switch and Math class)

```
import java.util.Scanner;

public class AreaCalculator {     public
static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Choose shape: 1.Circle  2.Rectangle  3.Triangle");
        int choice = sc.nextInt();
double area = 0;

        switch (choice) {
case 1:
            System.out.print("Enter radius: ");
double r = sc.nextDouble();              area =
Math.PI * r * r;
            break;

          case 2:
            System.out.print("Enter length and breadth: ");
double l = sc.nextDouble();              double b =
sc.nextDouble();              area = l * b;              break;

          case 3:
            System.out.print("Enter base and height: ");
double base = sc.nextDouble();              double h =
sc.nextDouble();              area = 0.5 * base * h;
            break;
```

```
            default:
                System.out.println("Invalid choice!");
                return;
        }

        System.out.printf("Area = %.2f\n", area);
        sc.close();
    }
}
```

Q Temperature Converter

```
import java.util.Scanner;

public class TemperatureConverter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("1. Celsius to Fahrenheit  2. Fahrenheit to Celsius");
int choice = sc.nextInt();

        System.out.print("Enter temperature: ");
double temp = sc.nextDouble();        double
converted = 0;

        converted = (choice == 1) ? (temp * 9 / 5) + 32 : (temp - 32) * 5 / 9;

        System.out.println("Converted Temperature: " + converted);
        sc.close();
    }
}
```

```
import java.util.Scanner;

public class NumberChecker {     public
static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
System.out.print("Enter a number: ");        int
num = sc.nextInt();

      // Even or Odd
if (num % 2 == 0)
         System.out.println(num + " is Even");
      else
         System.out.println(num + " is Odd");

      // Prime check
boolean prime = true;        if
(num <= 1) prime = false;
      else {
         for (int i = 2; i <= num / 2; i++) {
if (num % i == 0) {                  prime =
false;                  break;
            }
         }
      }
      System.out.println(num + (prime ? " is Prime" : " is Not Prime"));
```

Q Number Checker (Even/Odd, Prime, Palindrome)

```
      // Palindrome check
```

```
        int rev = 0, temp = num;
while (temp > 0) {            rev = rev
* 10 + temp % 10;            temp /=
10;
        }
        System.out.println(num + (num == rev ? " is Palindrome" : " is Not Palindrome"));


        sc.close();
    }
}
```

Q Pattern Printing

```
public class PatternPrinting {     public
static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Q Multiplication Table using do-while

```
import java.util.Scanner;

public class MultiplicationTable {
public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        int num;
        do {
            System.out.print("Enter a number (0 to exit): ");
num = sc.nextInt();

            if (num != 0) {                for
(int i = 1; i <= 10; i++) {
                    System.out.println(num + " x " + i + " = " + (num * i));
                }
            }
        } while (num != 0);
sc.close();
    }
}
```

Q 2D Fibonacci Table

```
import java.util.Scanner;

public class Fibonacci2D {     public
static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
System.out.print("Enter rows and columns: ");
int rows = sc.nextInt(), cols = sc.nextInt();         int[][]
fib = new int[rows][cols];

        int a = 0, b = 1, next;         for
(int i = 0; i < rows; i++) {
for (int j = 0; j < cols; j++) {
fib[i][j] = a;                 next = a + b;
a = b;
```

```
            b = next;
          }
        }

        System.out.println("\nFibonacci 2D Table:");
        for (int[] row : fib) {
for (int val : row)
            System.out.print(val + "\t");
          System.out.println();
        }
        sc.close();
    }
}
```

Matrix Operations (Addition Example)

```
import java.util.Scanner;

public class MatrixAddition {     public
static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
System.out.print("Enter rows and columns: ");        int
r = sc.nextInt(), c = sc.nextInt();

        int[][] A = new int[r][c];
int[][] B = new int[r][c];        int[][]
sum = new int[r][c];

        System.out.println("Enter Matrix A:");
        for (int i = 0; i < r; i++)
for (int j = 0; j < c; j++)
            A[i][j] = sc.nextInt();
```

```
        System.out.println("Enter Matrix B:");
        for (int i = 0; i < r; i++)
for (int j = 0; j < c; j++)
                B[i][j] = sc.nextInt();

        for (int i = 0; i < r; i++)
for (int j = 0; j < c; j++)
sum[i][j] = A[i][j] + B[i][j];

        System.out.println("Sum of Matrices:");
for (int[] row : sum) {            for (int val :
row)
                System.out.print(val + " ");
            System.out.println();
        }
        sc.close();
    }
}
```

Q Check if Two Strings Are Anagrams

Topic: String, Character Arrays, Sorting

```
import java.util.Arrays; import
java.util.Scanner;

public class AnagramChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
```

```
        String s1 = sc.nextLine().toLowerCase().replaceAll("\\s+", "");        System.out.print("Enter
second string: ");
        String s2 = sc.nextLine().toLowerCase().replaceAll("\\s+", "");

        if (s1.length() != s2.length()) {
            System.out.println("Not Anagrams!");
            return;
        }

        char[] arr1 = s1.toCharArray();
char[] arr2 = s2.toCharArray();
        Arrays.sort(arr1);
Arrays.sort(arr2);

        if (Arrays.equals(arr1, arr2))
            System.out.println("  Strings are Anagrams!");
        else
            System.out.println("  Strings are NOT Anagrams!");

        sc.close();
    }
}
```

Q Banking Transaction Log (Using StringBuilder)

Topic: StringBuilder, String search import java.util.Scanner;

```
public class TransactionLog {     public
static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        StringBuilder log = new StringBuilder();
```

```
        int choice;

        do {
            System.out.println("\n1. Add Transaction  2. Search  3. Show All  4. Exit");
System.out.print("Enter your choice: ");            choice = sc.nextInt();
sc.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    System.out.print("Enter Customer Name: ");
                    String name = sc.nextLine();
                    System.out.print("Enter Transaction Amount: ");
double amt = sc.nextDouble();                    log.append("Customer:
").append(name)
                      .append(", Amount: ").append(amt).append("\n");
                    break;

                case 2:
                    System.out.print("Enter Customer Name to Search: ");
                    String search = sc.nextLine();
                    String[] records = log.toString().split("\n");
boolean found = false;                    for (String r :
records) {                        if (r.contains(search)) {
System.out.println(r);                            found = true;
                        }
                    }
                    if (!found) System.out.println("No record found!");
                    break;
                case
3:
                    System.out.println("\nAll Transactions:\n" + log);
```

```
                break;
            case
4:
                System.out.println("Exiting...");
                break;
        }
    } while (choice != 4);

    sc.close();
  }
}
```

Q Word Frequency Counter

Topic: String Tokenizing import

```
java.util.*;

public class WordFrequency {    public
static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter a sentence:");
    String line = sc.nextLine();

    String[] words = line.toLowerCase().split("\\s+");
    Map<String, Integer> freq = new HashMap<>();

    for (String w : words) {          freq.put(w,
freq.getOrDefault(w, 0) + 1);
    }

    System.out.println("\nWord Frequency:");       for
(Map.Entry<String, Integer> e : freq.entrySet()) {
        System.out.println(e.getKey() + " : " + e.getValue());
```

```
        }

        sc.close();
    }
}
```

Q File Read/Write (Employee Data)

Topic: File I/O (Byte & Character Streams)

```
import java.io.*; import java.util.Scanner;

public class EmployeeFile {     public
static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        File file = new File("employee.txt");

        try (FileWriter fw = new FileWriter(file)) {
            System.out.println("Enter 3 employee records (ID Name Salary):");
            for (int i = 0; i < 3; i++) {                String id =
sc.next();                String name = sc.next();
double sal = sc.nextDouble();                fw.write(id
+ " " + name + " " + sal + "\n");
            }
        } catch (IOException e) {
            System.out.println("Error writing to file!");
        }

        System.out.println("\nReading data from file:");         try
(BufferedReader br = new BufferedReader(new FileReader(file))) {
            String line;            while ((line =
br.readLine()) != null) {
                System.out.println(line);
```

```
        }
      } catch (IOException e) {
        System.out.println("Error reading file!");
      }

      sc.close();
   }
}
```

Q Object Serialization and Deserialization

Topic: Serialization

```
import java.io.*; import
java.util.Scanner;

class Student implements Serializable {
String name;    int rollNo;
   double marks;

   Student(String n, int r, double m) {
name = n;       rollNo = r;       marks
= m;
   }

   void display() {
      System.out.println("Name: " + name + ", Roll: " + rollNo + ", Marks: " + marks);
   }
}

public class SerializeStudent {    public
static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Roll No: ");
        int roll = sc.nextInt();
        System.out.print("Enter Marks: ");
double marks = sc.nextDouble();

        Student s = new Student(name, roll, marks);

        // Serialization
        try (ObjectOutputStream oos = new ObjectOutputStream(new
FileOutputStream("student.ser"))) {
            oos.writeObject(s);
            System.out.println("  Student serialized to student.ser");
        } catch (IOException e) {
            System.out.println("Error in serialization!");
        }

        // Deserialization        try (ObjectInputStream ois = new ObjectInputStream(new
FileInputStream("student.ser"))) {
            Student obj = (Student) ois.readObject();
            System.out.println("\n--- Deserialized Student ---");
            obj.display();
        } catch (Exception e) {
            System.out.println("Error in deserialization!");
        }

        sc.close();
    }
}
```

16. Write a Java program to find the factorial of a number using a loop.

```
import java.util.Scanner; class Factorial
{    public static void main(String[] args)
{
      Scanner sc = new Scanner(System.in);
System.out.print("Enter number: ");
int n = sc.nextInt();        int fact = 1;
for (int i = 1; i <= n; i++) {           fact *= i;
      }
      System.out.println("Factorial = " + fact);
   }
}
```

17. Write a Java program to check if a string is a palindrome.

```
import java.util.Scanner; class
Palindrome {    public static void
main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter string: ");
      String str = sc.nextLine();        String
rev = "";        for (int i = str.length() - 1; i
>= 0; i--) {           rev += str.charAt(i);
      }
      if (str.equals(rev))
         System.out.println("Palindrome");
      else
         System.out.println("Not Palindrome");
   }
}
```

18. Write a Java program to count vowels and consonants in a string.

```
import java.util.Scanner; class
CountVowels {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter string: ");
String s = sc.nextLine().toLowerCase();
      int v = 0, c = 0;        for (char
ch : s.toCharArray()) {            if
("aeiou".indexOf(ch) != -1)
            v++;           else if (ch >=
'a' && ch <= 'z')
            c++;
      }
      System.out.println("Vowels: " + v + ", Consonants: " + c);
   }
}
```

19. Write a Java program to find the largest element in an array.

```
class LargestInArray {     public static
void main(String[] args) {
      int[] arr = {5, 10, 2, 8, 15};
int max = arr[0];        for (int i = 1; i
< arr.length; i++) {            if (arr[i] >
max)               max = arr[i];
      }
      System.out.println("Largest element: " + max);
   }
}
```

20. Write a Java program to reverse an array.

```
class ReverseArray {     public static void
main(String[] args) {        int[] arr = {1, 2,
3, 4, 5};
      System.out.print("Reversed Array: ");
for (int i = arr.length - 1; i >= 0; i--) {
         System.out.print(arr[i] + " ");
      }
   }
}
```

---

21. Write a Java program to find the sum of digits of a number.

```
import java.util.Scanner; class
SumOfDigits {     public static void
main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter number: ");
int num = sc.nextInt();        int sum = 0;
while (num > 0) {            sum += num %
10;           num /= 10;
      }
      System.out.println("Sum of digits = " + sum);
   }
}
```

---

22. Write a Java program to check whether a number is prime or not.

```
import java.util.Scanner; class
PrimeCheck {
   public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
System.out.print("Enter number: ");
int n = sc.nextInt();       boolean prime =
true;       if (n <= 1) prime = false;
    else {          for (int i = 2; i <=
n / 2; i++) {               if (n % i == 0)
{                   prime = false;
break;
            }
        }
    }
    System.out.println(prime ? "Prime Number" : "Not Prime");
  }
}
```

---

23. Write a Java program to generate the Fibonacci series.

```
import java.util.Scanner; class
Fibonacci {    public static void
main(String[] args) {
    Scanner sc = new Scanner(System.in);
System.out.print("Enter count: ");
    int n = sc.nextInt();
int a = 0, b = 1;
    System.out.print("Fibonacci: " + a + " " + b + " ");
for (int i = 2; i < n; i++) {          int c = a + b;
        System.out.print(c + " ");
        a = b;
b = c;
    }
  }
}
```

24. Write a Java program to display multiplication table of a number.

```
import java.util.Scanner; class
MultiplicationTable {     public static
void main(String[] args) {
      Scanner sc = new Scanner(System.in);
System.out.print("Enter number: ");
int n = sc.nextInt();        for (int i = 1; i <=
10; i++) {
          System.out.println(n + " x " + i + " = " + (n * i));
      }
   }
}
```

25. Write a Java program to swap two numbers without using a third variable.

```
import java.util.Scanner; class
SwapNumbers {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
System.out.print("Enter first number: ");
      int a = sc.nextInt();
      System.out.print("Enter second number: ");
      int b = sc.nextInt();
a = a + b;        b = a - b;
a = a - b;
      System.out.println("After swap: a = " + a + ", b = " + b);
   }
}
```